Computational Programming with Python

Lecture 3: String Formatting, other Container Types and Floating Point Numbers

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This lecture

- Training exercises
- String formatting
- Dictionaries
- Sets
- Tuples
- Container conversions
- Floating point numbers

Basic string formatting

Book p. 43

- Using keywords:

In [ ]:

```
str1 = "I'm a {somethingelse}.".format(somethingelse = "string")
str2 = "I'm a {something}.".format(something = 10)
something = 10
str3 = f"I'm a {something}."
print(str2)
print(str3)
```

- Using positional arguments:

In [ ]:

```
print("Two strings, {} and {}.".format("a", "b"))
```

f-strings

An elegant alternative in modern Python is the use of `f-strings`.

In [ ]:

```
quantity = 33.45
other_quantity = 2
text = f"""
We use {quantity} g sugar and {other_quantity} g salt.
In total {quantity + other_quantity} g."""
print(text)
```

To enable this interpretation of the text enclosed in `{...}` the string has to prefixed by the letter `f`.

Old string formatting

- for strings

In [ ]:

```
course_code = "NUMA01"
print("This course’s code is %s" % course_code)
```

- for integers

In [ ]:

```
nb_students = 16
print("There are %i students" % nb_students)
```

- for floats

In [ ]:

```
average_grade = 3.4
print("Average grade: %f" % average_grade)
```

See for example **C function printf** for explanation of formatting characters.

Formatting numbers using f-strings

We can right-align numbers at some position which is written after a colon.

In [ ]:

```
print("number      square")
print("------------------")
for i in range(-2, 3):
    print(f"{i:6} {i*i:10}")
    #print(f"{i} {i*i}")
```

Alignment and decimals

We can add `.2f` to show two decimals:

In [ ]:

```
from numpy import arange
print("number      square")
print("------------------")
for i in arange(0.2, 0.5, 0.1):
    print(f"{i:6.2f} {i*i:10.2f}")
```

Decimals and scientific notation

More examples

In [ ]:

```
quantity = 33.45
print(f"five decimals: {quantity:0.5f}")
print(f"scientific notation with two decimals: {quantity:0.2e}")
```

Dictionaries

A dictionary is an unordered structure of `key:value` pairs. It is similar to a list but the objects are accessed by `keys` instead of index.
One indicates dictionaries by curly brackets:

In [ ]:

```
homework = {'Alice': True, 'Bob': False}
homework['Alice']
```

In [ ]:

```
homework['Bob'] = True
homework['Bob']
```

Adding and deleting

In [ ]:

```
homework = {'Alice': True, 'Bob': False, 'Alice': False}
homework['Caesar'] = True
print(homework)

del homework['Bob']
print(homework)
```

Note: The keys must be **unique**.

Datatypes of keys

Any **immutable** (unchangable) object can be used as a key, e.g. strings, numbers, Booleans, etc are fine, but **not lists**.

In [ ]:

```
my_dict = {1: "alpha", 2: "beta", 3: "gamma", "lang": "Greek", 5: [1,2,3]}
print(my_dict[1])
print(my_dict[5])
print(my_dict["lang"])
aa = 10
my_dict[ aa ] = True
print(my_dict)
```

Looping through dictionaries

A dictionary is an object with the following useful methods: `keys`, `items`, `values`.

By default a dictionary is considered as a list of keys:

In [ ]:

```
for key in homework:  # or homework.keys()
    print(f"The key {key} has the value {homework[key]}.")
```

In [ ]:

```
for key, value in homework.items():  # using the items method
    print(f"{key}: {value}")
```

In [ ]:

```
for value in homework.values():  # using the value method
    print(value)
```

Dictionaries in this course

Dictionaries are mainly used for

- providing functions with argument in a compact way (Lecture 4)
- to collect options of a method:

```
{'tol':1.e-3, 'step':0.1, 'maxit':1000}
```

Sets

Sets are containers which mimic sets in their mathematical sense:

Definition:

A set is a collection of well defined and distinct objects, considered as an object in its own right. (Wikipedia)

In [ ]:

```
fruits = set(['apple', 'pear', 'banana', 'banana'])
print(fruits)
```

The most important operation is `in`, meaning $\in$ (is element of):

In [ ]:

```
'plum' in fruits
```

In [ ]:

```
'pear' in fruits
```

Operations on Sets

- $A \cap B$ - method `intersection`
- $A \cup B$ - method `union`
- A/B, relative complement - subtract operation

In [ ]:

```
fruitbasket = set(['apple', 'pear', 'banana', 'banana'])
print(fruitbasket)
rotten = set(['pear', 'orange'])
exotic = set(['banana', 'mango', 'kiwi'])
emptyset = set([])

print(emptyset == fruitbasket.intersection(rotten)) # are all edible

extendedbasket = fruitbasket.union(exotic)
goodfruits = fruitbasket - rotten

print("extendedbasket =", extendedbasket)
print("goodfruits =", goodfruits)
```

No duplicate elements

"...distinct objects..." (see mathematical definition)
This is reflected in Python by

In [ ]:

```
s = {'apple', 'apple', 'pear'}
print(s)
set(['apple', 'apple', 'pear']) == set(['apple', 'pear'])
```

Tuples

Definition

A **tuple** is an **immutable** list. Immutable means that it cannot be modified.

Example

In [ ]:

```
my_tuple = (1, 2, 3) # our first tuple
my_tuple = 1, 2, 3   # parentheses not required
print("The length of my_tuple is", len(my_tuple))  # same as for lists

# my_tuple[0] = 'a'    # error! tuples are immutable
my_tuple = ('a', 2, 3)

singleton = 1,    # note the comma
singleton = (1,)  # preferred for readability
print(singleton[0])
print(type(singleton))
```

Packing and unpacking

One may assign several variables at once by **unpacking** a list or a tuple:

```
a, b = 0, 1      # a gets value 0 and b gets value 1
a, b = [0, 1]    # exactly the same
(a, b) = 0, 1    # same
[a, b] = [0, 1] # same
```

The swapping trick!

```
a, b = b, a
```

A final word on tuples

- Tuples are nothing else than immutable lists
- In most cases lists may be used instead of tuples
- The parentheses free notation is nice but **dangerous**, you should **use parenthesis when you're not sure**:

```
a, b = b, a    # the swap trick, equivalent to
(a, b) = (b, a)
# but:
```

In [ ]:

```
t1 = a, b = [0, 1]
print(type(t1))

value1 = 1, 2 == 3, 4        # a tuple
value2 = (1, 2) == (3, 4)    # a bool
print("value1 =", value1)
print("value2 =", value2)
```

Container conversions

From	To	Command
List	Tuple	`tuple([1, 2, 3])`
Tuple	List	`list((1, 2, 3))`
List, Tuple	Set	`set([1,2,3]), set((1,2,3))`
Set	List	`list({1, 2, 3})`
Dictionary	List	`{'a':4}.values()`
List	Dictionary	-

Summary and overview

Type	Access	Order	Duplicate values	Mutability
List	index	yes	yes	yes
Tuple	index	yes	yes	no
Dictionary	key	no	yes	yes
Set	no	no	no	yes

What are Floating Point Numbers?

Floating point numbers – The number e

The number e can be defined as

$$e = e^1 = \exp(1) = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n.$$

In [ ]:

```
from numpy import *

e = exp(1)
print(f"{e:2.16f}")

n = 100000000
e = (1 + 1/n)**n
print(f"{e:2.16f}")
```

In [ ]:

```
n = 1000000000000000000
e = (1 + 1/n)**n
print(f"{e:2.16f}")
```

How to create a Floating Point Number?

- By typing it:

In [ ]:

```
a = 3.012
b = 30.12e-1
c = 0.3012e+1
d = 3e-1
print(a)
print(b)
print(c)
print(d)
```

- As a result from a computation

In [ ]:

```
n = 2        # This is an integer
a = n/n      # This is a float
m = int(n/n)
print(n)
print(a)
print(m)
```

Comparing floating point numbers

Operations on floating point numbers rarely return the exact result:

In [ ]:

```
a = 0.4 - 0.3
print(f"{a:2.18f}")
```

This fact matters when **comparing** floating point numbers:

In [ ]:

```
(0.4 - 0.3) == 0.1
```

Never compare floats using `==`! Ask instead if they are `near`:

In [ ]:

```
abs((0.4 - 0.3) - 0.1) < 1e-13
```

In [ ]:

```
from numpy import exp, inf, nan # not a number
#print(exp(1000.))
a = inf
print(3 - a)
print(3 + a)
print(a + a)
print(a - a)
print(a / a)
```

Example for internal representation (simplified)

Roughly speaking, a floating-point number is represented by a fixed number of significant digits (the significand) and scaled using an exponent in some fixed base.
The base for the scaling is normally 2, 10 or 16.
A number that can be represented exactly is of the following form:

$$\text{significand} \times \text{base}^{\text{exponent}},$$

where `significand`, `base` and `exponent` are integers (which have a simple representation).
Note: The significand is also called `mantissa`.

Example:

$$1.2345 = \underbrace{12345}_{\text{significand}} \times \underbrace{10}_{\text{base}}{}^{\overbrace{-4}^{\text{exponent}}}$$

Internal representation

Floats are represented by three quantities: The *sign*, the *significand* (or *mantissa*) and the *exponent*:

$$\operatorname{sign}(x)\left(x_0 + x_1\beta^{-1} + \cdots + x_{t-1}\beta^{-(t-1)}\right)\beta^k, \quad \beta \in \mathbb{N}, \quad x_0 \neq 0, \quad 0 \leq x_i < \beta.$$

- $x_0, \ldots, x_{t-1}$ is the significand,
- t is the length of the significand,
- β is the basis,
- k is the exponent with $|k| \leq k_{max}$.

Normalization $x_0 \neq 0$ makes the representation unique (and saves one bit in the case $\beta = 2$ and allows to add $x_t\beta^{-t}$).
On your computer for a normal float (double):

- $\beta = 2$,
- $t = 52$,
- $k_{max} = 1023$ (which needs 10 bits).

Together with the sign and the full range (sign) of the exponent, this requires 64 bits.

Internal representation (cont.)

We then have the following representation (adding $x_t\beta^{-t}$ because x_0 is fixed):

$$(-1)^{\text{sign}}\left(\underbrace{1}_{=x_0}+\sum_{i=1}^{t}x_i2^{-i}\right)2^k, \quad x_i\in\{0,1\}.$$

0 cannot be represented as a *normalized* float.
Two zeroes: All the exponent bits 0 with all significand bits 0 represents 0. If sign bit is 0, then we have 0, else -0.
Infinity: All the exponent bits 1 with all significand bits 0 represents infinity.
Denormalized number: All the exponent bits 0 and significand bits non-zero represents denormalized number.
Error: All the exponent bits 1 and significand bits non-zero represents error or `nan` (Not a Number).

Precision – Machine epsilon

Definition (Maching epsilon):

The *machine epsilon* or rounding unit is the gap between 1.0 and the next larger floating point number, i.e. the largest number ϵ such that

$$\mathrm{float}(1.0 + \epsilon) = 1.0$$

Challenge: Write a program which finds this number (or a good estimate for it).
See also

In [ ]:

```
import sys
sys.float_info.epsilon
```

What precision do I have at what number

What decimal precision do I have at a given number $a \in \mathbb{R}$?
To figure out the precision we have at a number, we find the interval such that $a \in [2^k, 2^{k+1})$. We then subdivide that range using the `significand` bits.
Example: $3.5 \in [2, 4)$. A float in Python has 52 bits of `significand`, so the precision we have at 3.5 is:

$$(4-2)2^{-52} = \frac{2}{4503599627370496} \approx 0.000000000000000044408921$$

3.5 itself is actually exactly representable by a float, but the amount of precision numbers we have at that scale is that value. The smallest number you can add or subtract to a value between 2 and 4 is that value. That is the resolution of the values you are working with when working between 2 and 4 using a float.

In [ ]:

```
prec = (4-2) * 2**-52
print(f"prec for 3.5 is {prec}")
```

1. What decimal precision do I have at 13689.0?
2. What is the largest possible floating point number in this double precision representation?
3. What is the smallest gap between floating point numbers around zero?

5min, let's go!

In [ ]:

```

```

In [ ]:

```
prec = (2**14 - 2**13)* 2**-52
print(f"prec for 13689 is {prec}")

largest = 2**1023
print(f"Largest floating point number: {largest:2.4e}")

smallest = 2**-1023
print(f"Distance between 0 and next number: {smallest:2.4e}")
```

Internal representation (cont.)

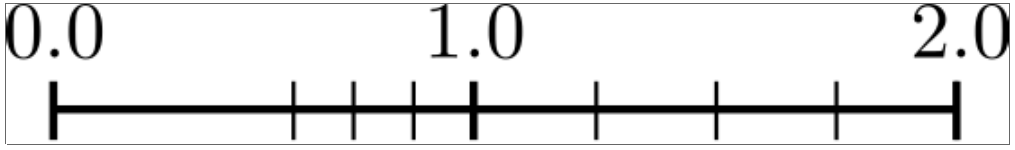
The smallest positive representable number is

$$\mathrm{fl}_{\min} = 1 \cdot 2^{-1023} \approx 10^{-308}$$

and the largest is

$$\mathrm{fl}_{\max} = 1.111\ldots1 \cdot 2^{1023} \approx 10^{308}.$$

Floating point numbers are not equally spaced in $[0, \mathrm{fl}_{\max}]$.
Gap at zero (caused by normalization $x_0 \neq 0$):
Distance between 0 and the first positive number is 2^{-1023}.
Distance between the first and the second is smaller by a factor 2^{-52}.



Infinite and Not a Number

There are in total $2(\beta-1)\beta^{t-1}(2k_{max}+1)+1$ floating point numbers, in short floating point numbers are limited.
Numbers bigger than the maximal floating point number $\mathrm{fl}_{\max}$ generate in numpy an `inf` or an overflow exception.
Numbers smaller than the minimal floating point number $\mathrm{fl}_{\min}$ generate subnormalized numbers or `not a number`, short `nan`. This leads to unexpected behavior.

In [ ]:

```
x = nan
print(x < 0)
print(x > 0)
print(x == x)
```

The float `inf` behaves much more as expected:

In [ ]:

```
print(0 < inf)
print(inf <= inf)
print(inf == inf)
print(-inf < inf)
print(inf - inf)
print(exp(-inf))
print(exp(1/inf))
```